NYU Stern School of Business
Center for the Future of Management
44 W 4th St
New York, NY 10012

Faisal D'Souza, NCO
Office of Science and Technology Policy
Executive Office of the President
2415 Eisenhower Avenue
Alexandria, VA 22314

Introduction

We are a group of economic, policy, and AI experts coordinated by the New York University Stern School of Business Center for the Future of Management and are grateful for the opportunity to provide input to the White House Office of Science and Technology Policy's request for information on the Artificial Intelligence Action Plan. We see AI as a critical technology for the future of our country, and we see policy as playing a pivotal role in ensuring this future.

AI Policy

Artificial intelligence is a critical technology and leadership in AI is a key to America's success. Fortunately, policymakers appreciate the importance of this emerging technology, and it has been a central focus for both the White House and U.S. Congress over the past several years. The first Trump, Biden, and second Trump Administrations have all issued separate executive orders on AI. Last year, the Bipartisan Senate AI Working Group released a roadmap for AI policy in May and the Bipartisan House Task Force on AI released a report on AI in December. While there are disagreements between the political parties on many issues, AI has seen considerable amounts of bipartisanship in Washington. We believe AI should and can remain a bipartisan issue, even if the policy priorities change from administration to administration.

The Trump Administration's RFI has highlighted two key policy priorities for the new administration:

- The desire to "sustain and enhance America's AI dominance"
- Ensuring "that unnecessarily burdensome requirements do not hamper private sector AI innovation"

We believe promoting U.S. competitiveness and innovation through smart regulation and policy are important goals. We would like to offer our expertise in four key topics that we think are critical to achieving these goals:

1. Productivity and Competitiveness
2. Innovation
3. Preparing the Workforce for Job Disruption
4. Regulation

Productivity and Competitiveness

AI's potential to boost productivity is critical to U.S. competitiveness. Historically, productivity gains have been a major driver of rising standards of living. The U.S.'s ability to tap into this potential can help critical sectors like technology, manufacturing, energy, financial services, and healthcare outperform international competitors.

Understanding how the technology will impact firm and worker productivity is critical to tapping into AI as a source of American competitiveness. Economists have conducted firm level studies, finding that firms which invest more in AI experience higher growth in sales, employment, and market valuations. A literature review analyzed AI's impact on worker productivity and found preliminary evidence suggesting that AI generally boosted worker productivity in areas analyzed by economists, such as coding and writing. The review also found limits and barriers to AI boosting productivity, including some areas where AI hurt productivity.

The literature review further highlighted some limits to the existing studies: 1) The studies only looked at a small number of tasks workers completed in the economy; 2) Many of the studies were in more experimental rather than real world settings; 3) The studies were short-term in nature; 4) The studies did not look into how productivity might dramatically change as firms restructure around a technology and new firms and tasks emerge.

Other studies have discussed several examples of AI, as a general-purpose technology, that harbors the potential to significantly enhance productivity and address some of the most pressing global challenges. For example, generative AI is showing the promise of increased productivity for management consultants both with respect to quantity of tasks completed (12.2 percent more tasks than a control group that did not use generative AI) and the quality achieved (40 percent higher). As another example, AI-based chatbots have been shown to increase productivity of customer service agents, with the greatest impact on novice and low-skilled workers.

The impacts of productivity on government efficiency and business competitiveness are critical to America's success. Policymakers can help address some of the limitations of these studies through a series of pilot programs in the federal government and grants to corporations, universities, and other organizations in the U.S. to better understand how AI impacts American worker productivity and competitiveness over the short- and long-term. The research can help identify the strengths, limits, and best practices for leveraging the technology to boost U.S. productivity and competitiveness.

Innovation

AI has considerable potential to boost scientific discovery and innovation. AI can advance science in fields ranging from energy to semiconductors to pharmaceuticals by automating and augmenting work done by scientists, engineers, and other researchers.

Quality economic studies analyzing AI's impact on scientific discovery in a real world setting are currently sparse. Fortunately, a recent working paper from MIT analyzed the randomized introduction of an AI tool in a corporate lab in the field of materials sciences. The research found that scientists using the tool saw a 44% increase in materials discovery, 39% increase in patent filings, and 17% rise in downstream production innovation. However, the study also showed that a majority of these scientists reported higher levels of job dissatisfaction over the two year period the AI tool was introduced.

This research suggests AI could play a major role in advancing scientific discovery, which can foster major innovation and boost United States' technological and economic competitiveness. The study also shows some of the challenges that might be encountered in doing so that policymakers can help address.

The potential for AI to accelerate discovery in science suggests that policymakers should take a holistic approach to AI, R&D, and innovation by looking across scientific disciplines. R&D in AI can boost innovation in AI, but by itself is unlikely to tap into the full potential of AI to boost innovation in key industries like energy, manufacturing, and healthcare. The successful adoption of AI in ways that boost innovation can help American scientists build familiarity and skills with the technology in ways that can further drive American innovation.

The administration could support increased funding across all domains that can benefit from leveraging AI for R&D or focus on key areas they want to prioritize, such as energy and biotechnology. Increased funding can help achieve other priorities set by the Trump Administration, such as "energy dominance." For instance, new materials discovery resulting from the use of AI, could be a key to improving our energy infrastructure and production in ways that enable us to lead in this field and support other industries through lower energy costs.

The administration can also create pilot programs and funding to help identify opportunities, bottlenecks, and limitations to the use of AI for scientific discovery to help efficiently allocate resources and address challenges. For instance, further research can identify whether the job dissatisfaction among scientists uncovered by the MIT study at the materials science lab is common across other domains and how this challenge might be addressed.

The competitiveness of the U.S. economy will be significantly boosted if our federal labs, universities, and corporations realize how to best leverage AI to foster innovation across different scientific fields and industries.

Preparing the Workforce for Job Disruption

As some have argued, in order to succeed in an AI-first world, we need to create an AI-ready workforce. While the impact of AI on jobs is hard to predict, we see opportunities and challenges that AI might bring to workers. For instance, we see the potential for AI to create new industries that create the jobs of the future. However, we also see challenges with AI potentially automating many tasks currently done by entry level workers that serve as a way for these workers to enter a profession and build the valuable experience and connections they need to advance.

If history is a guide, automation's impact on net jobs is unpredictable, even in industries where the automation is happening. For instance, automation corresponded with a decline in the share of jobs in the agriculture sector, but (counterintuitively) an increase in the share of jobs in the automotive sector after the rise of the assembly line.

There are some key differences between AI and past waves of automation. Unlike the industrial and information technology revolutions that focused on automating routine tasks following an explicit set of instructions (e.g. a sewing machine moving a needle up and down in a pattern or a spreadsheet sorting files by date) AI is focusing on non-routine tasks that do not rely on explicit instructions, such as drawing an image, preparing a presentation, or writing a first draft of an article.

Economists have tried to take a forward looking approach by identifying occupations and sectors of the economy that are relatively exposed to advances in AI. Results from these approaches highlight that white collar workers are likely more exposed to advances in AI, but that there will likely be lots of variation across workers. Some of these approaches conceive of jobs as a series of tasks, and ask for each of these tasks whether AI can substitute for a human in completing the task. Jobs that have more tasks for which AI can substitute for a human are considered more exposed.

Economists are only just beginning to understand conditions under which AI automates and augments human jobs. Just because a job may be highly exposed to advances in AI does not necessarily imply that the job is at risk of automation by AI. For example, in some cases even if AI "could" substitute for a human for some tasks, the relative costs of substitution might be too high. Instead, the nature of the job itself might change in such a way that a human continues to do some tasks associated with the job whereas AI does other of the tasks.

As highlighted in a recent National Academies report on artificial intelligence and the future of work, there are still many unknowns about how the technology will evolve, and about how it will impact the economy and workforce. Given that the impact AI will have on jobs will be hard to predict, we suggest the following approaches:

> 1) upgrading the federal statistical system to meet the research and policymaking needs created by AI;
>
> 2) filling data gaps to help policymakers and economists better monitor emerging trends;
>
> 3) and supporting policies that promote worker resilience and adaptability given the uncertainties around AI's impact on labor markets.

First, as a recent American Statistical Association report recognizes, greater investments in the federal statistical system are necessary to upgrade the nation's statistical infrastructure and meet the nation's evolving needs for timely, trusted, and unbiased information. Indeed, the federal statistical agencies have seen their budgets erode in inflation-adjusted terms over the last 15 years (whereas federal discretionary, nondefense spending has increased in real terms). The necessary investments extend beyond budgetary resources, suggesting that progress can be made on many different fronts, including removing barriers to innovation in the federal statistical system and protecting the statistical agencies from undue political interference.

Second, echoing a recent American Economic Association Committee on Economic Statistics white paper, we need better data to monitor the situation and have several data gaps that we need to fill. Since 2018, the U.S. Census Bureau has included questions about AI in its Annual Business Survey (ABS), and since December 2023 it has been tracking the adoption of AI in its bi-weekly Business Trends and Outlook Survey (BTOS). An advantage of the BTOS data is that it is up-to-date. However, a limitation is that it is collected at the company level rather than at individual business locations. Large companies often have multiple locations spread across different regions, which makes it difficult to analyze local impacts. To understand how AI adoption affects workers and local economies, we need data at the level of individual business locations.

Moreover, many important policy questions cannot be fully addressed without such detailed data. For example, understanding how technology changes workers' experiences is crucial for shaping effective education, job training, and retraining programs. Without this insight, it becomes challenging to prepare individuals for the evolving workforce. Insight and best practices for preparation can be derived from the fields that have been most heavily and immediately impacted, such as writers.

There are also unanswered questions about how AI adoption affects different regional economies. In some areas, the impact on jobs may be more severe because industries—and even specific job roles—are often concentrated in certain locations. If AI and robotics replace jobs in these industries, regions that depend on them for employment and tax revenue could face significant economic challenges. Indeed, nascent research suggests that AI adoption and robot adoption are highly geographically concentrated.

Further, collecting and sharing detailed data enables more accurate analysis of these complex issues. It also allows multiple researchers to verify findings, strengthening the reliability of studies and supporting better decision-making.

Second, the uncertainty around AI's impact on jobs requires a policy approach that fosters worker resilience and adaptability. One way that organizations can support this is with activities such as the GenAI promptathons which some have studied. Lifelong learning should be a key pillar to this approach. Policymakers can consider ways to both finance and develop an infrastructure around lifelong learning, such as using lifelong learning accounts, apprenticeship programs, and innovative credentialing methods.

Another key pillar should be promoting AI literacy. Ensuring workers understand the opportunities and limits of AI can help prepare them to leverage the technology and remain competitive in the labor market. The recent National Academies report on artificial intelligence and the future of work highlighted that AI holds promise as a tool to help train the workforce. An empirical evaluation of providing a health system's workforce access to a secure, HIPAA-compliant GenAI model and support for its use is one example. Other areas, such as workplace governance around AI tools and social safety net design, can also be key pillars to this goal.

Regulation

The regulatory framework around AI has been evolving over the years domestically and internationally. Many federal agencies have considered ways to update their regulations and guidelines for AI, while state governments have been looking into new legislation. Regulation is not inherently good or bad. When done well, it can help address concerns to build trust in a technology and foster adoption, but when done poorly, it can stifle innovation through excessive costs, restrictions, and unintended consequences. We offer the following observations to help guide the administration's approach to regulation.

The National Institute of Standards and Technology has taken a multistakeholder approach to designing voluntary soft law around AI, such as its AI risk management framework. While different administrations can and should decide what priorities to set for NIST, we strongly encourage NIST's work on AI be properly funded to carry out its important work. John Soroushian (one of the authors) has found NIST to be very popular among various stakeholders he has interacted with from industry, civil society, academia, and government, including both Republicans and Democrats, throughout his work on AI policy in Washington. Soft law approaches like those provided by NIST can be flexible and serve as a model for hard law approaches, if the executive branch and Congress decide hard law is appropriate. An iterative approach to soft law can help foster learning and ensure that any hard law approaches modeled after it are well-designed and not overly burdensome, or even address certain concerns in a way that hard law may not be necessary.

Leveraging existing laws when appropriate and tailoring regulation based on risk is also popular and can help reduce regulatory burdens. While working on the national strategy on AI for Congress and in his subsequent work, Soroushian has seen support for these two principles. He has further seen the House continue with this thinking in the Bipartisan Task Force on AI, which noted that "Policymakers can avoid duplicative mandates if they consider whether issues raised by AI are truly novel and without precedent or if existing laws and regulations already address the underlying concern" and which further stated "A thoughtful, risk-based approach to AI governance can promote innovation rather than stifle it."

Concluding Thoughts and Further Engagement

AI's transformative potential requires a whole-of-government approach that balances various policy goals. The administration's work on AI is critical and we appreciate the opportunity to comment on this key matter. We are happy to further elaborate on the issues discussed in our comment letter or other issues within our domain of expertise.

Author Bios

John Soroushian is a fellow at NYU Stern School of Business. He has previously worked at the Bipartisan Policy Center, US Treasury Department, and Brookings Institution. At the Bipartisan Policy Center, he initiated the think tank's work on AI and co-led an initiative to help put together a national strategy on AI for Congress that became the basis for a House Resolution that passed in 2020. This think tank effort took place in partnership with Reps. Will Hurd and Robin Kelly and involved collaboration with key policymakers and stakeholders, including key members of the Trump Administration, such as Dr. Lynne Parker. John has led multiple technology policy initiatives since and has briefed members of Congress and the National AI Advisory Committee on AI policy.

Dr. Tania Babina is an Associate Professor at the Robert H. Smith School of Business at the University of Maryland. She is also an affiliate of the National Bureau of Economic Research (NBER) and a Research Fellow at the Centre for Economic Policy Research (CEPR). Her work has been published in top academic journals and has been cited in the Financial Times, Washington Post, and The New York Times, among others. Her research was cited in major recent US regulations on Open Banking and in the government assessment of the UK Open Banking regulation as well as in the US White House Economic Report of the President.

Dr. Erik Brynjolfsson is a Professor at Stanford with appointments in Economics, Business and the Institute for Human-center Artificial Intelligence. He directs the Stanford Digital Economy Lab and co-chaired the committee on AI and the Future of Work for the National Academy of Sciences, Engineering and Medicine. Brynjolfsson is the co-author of nine books including *The Second Machine Age* and is one of the most cited researchers studying the productivity and workforce implications of AI.

Dr. Avinash Collis is an Assistant Professor at the Heinz College at Carnegie Mellon University. His research interests include the economics of digitization, focusing on measuring the welfare gains from digital goods. His research has been covered in major media outlets and policy reports including the *New York Times*, *Wall Street Journal*, *Washington Post*, the *Economist*, *CNN*, *BBC*, *Financial Times*, *Bloomberg*, and *NPR*, and reports by the U.S. White House, Federal Reserve, Senate, and UK treasury.

Dr. Christophe Combemale is an assistant research professor at Carnegie Mellon University in the Department of Engineering and Public Policy, with an affiliate appointment at the Heinz College of Information Systems and Public Policy. He is CEO and Principal Partner at Valdos Consulting LLC. He is the co-lead of the Workforce Supply Chains Initiative at the Block Center for Technology and Society, which seeks to develop new tools for understanding the readiness of labor markets to respond to new skill demand from industrial expansion, and transition options for workers who face employment disruption. He has advised the National AI Advisory Committee, US DoD, US Department of Commerce and state governments about workforce implications of AI, the possible patterns of automation for different forms of AI technology, and the potential resilience of labor markets to technological disruption.

Dr. Jane Dokko was the Chief Economist of the Department of Commerce during 2023-2024, where she led critical analysis of a wide range of domestic issues related to U.S. competitiveness, innovation, and economic growth. She works at the intersection of public policy and research, examining the most important forces shaping economic opportunity and inequality. Her research has been published in leading academic journals and widely cited in the media, including *The Wall Street Journal, Bloomberg, The Washington Post*, and others.

Dr. Martin Fleming is a Research Scientist at MIT Sloan and engaged in MIT's Computer Science and Artificial Intelligence Lab (CSAIL) FutureTech Project at the intersection of economics and computer science. Martin is the former IBM Chief Economist and Chief Analytics Officer and former Chief Revenue Scientist, Varicent, the Toronto-based sales-performance management software provider. He is a Fellow of The Productivity Institute, the UK research organization exploring what productivity means for business, workers, and communities and a consultant to the U.S. Bureau of Economic Analysis.

Dr. Anindya Ghose is the Heinz Riehl Chair Professor of Technology and Marketing at New York University's Leonard N. Stern School of Business. He is the author of two best selling books, *TAP: Unlocking The Mobile Economy* and *THRIVE: Maximizing Well-Being in the Age of AI*. He has worked on digital platforms, reputation and rating systems, digital marketing, data privacy, mobile economy and AI. His research has been published in leading academic journals and been cited in numerous mainstream outlets including *The New York Times, The Wall Street Journal, CNN, BBC* and others.

Dr. Osea Giuntella is an Associate Professor of Economics at the University of Pittsburgh and a Research Associate with the National Bureau of Economic Research (NBER). His research focuses on health economics and public policy, particularly the impact of technology, robotics, and AI on worker health, as well as the role of policies and institutions. His work has been published in top-tier academic journals, has informed policymakers, and received media attention. In 2024, he was invited to speak at the EU Council's Ministerial Conference on worker well-being in the age of AI.

Dr. Yong Lee is an Associate Professor of Technology, Economy, and Global Affairs at the Keough School of Global Affairs at the University of Notre Dame. Lee is also the Technology Ethics Program Chair at the Institute for Ethics and the Common Good. His research examines technology and labor, focusing on how artificial intelligence and robotics affect labor and inequality and how society interacts with these new technologies.

Sam Manning is a Senior Research Fellow at the Centre for the Governance of AI (GovAI). His work focuses on measuring the economic impacts of frontier AI systems and designing policy options to help ensure that advanced AI can foster broadly shared economic prosperity. He previously conducted research at OpenAI and worked on a randomized controlled trial of a guaranteed income program in the US. Sam has a MSc in International and Development Economics from the University of San Francisco.

Dr. Alexander Oettl is Professor of Strategy and Innovation at the Georgia Tech Scheller College of Business and a Research Associate at the National Bureau of Economic Research. His research focuses on the economics of innovation, entrepreneurship, and science. His research has been profiled in numerous international media outlets including the *Wall Street Journal*, *New York Times*, *Financial Times*, *Harvard Business Review*, *Nature*, *Science*, *Wired*, among others.

Dr. Manav Raj is an Assistant Professor in the Management Department at the Wharton School of the University of Pennsylvania. His research primarily focuses on understanding the downstream implications of technological change on competition and strategy, with a focus on digital technologies and artificial intelligence. Further, he studies how institutional features and non-market forces can shape innovation.

Dr. Daniel Rock is an Assistant Professor in the Operations, Information, and Decisions department at the Wharton School of the University of Pennsylvania. His research focuses on the economic impact of digital technologies, with a particular focus on artificial intelligence. He is also a Digital Fellow at both Stanford HAI's Digital Economy Lab and the MIT Initiative on the Digital Economy, and a Schmidt Futures AI2050 Early Career Fellow. He received his B.S. in Economics from the University of Pennsylvania, and S.M. and Ph.D. from MIT.

Dr. Robert Seamans is a Professor at New York University’s Stern School of Business, Director of the NYU Stern Center for the Future of Management, and a nonresident Senior Fellow at the Brookings Institution. His research focuses on the economic consequences of AI, robotics and other advanced technologies. His research has been published in leading academic journals and has been widely cited. In 2015, Professor

Seamans was appointed as the Senior Economist for technology and innovation on the White House Council of Economic Advisers.

Dr. Neil Thompson is the Director of MIT FutureTech, a group of economists and computer scientists hosted in MIT's Computer Science and AI Lab and the MIT Sloan School of Management. His work focuses on how AI is evolving and what that means for job automation and scientific progress. He has been published in top academic journals across five fields and written about in the *New York Times*, *Time* and many others.

Dr. Batia Wiesenfeld is the Andre J.L. Koo Professor of Management at New York University's Stern School of Business, Director of the Business & Society Program, and has an affiliated appointment in the Department of Population Health at NYU Langone Health. She is an expert on organizational change and her recent NSF- and NIH-supported research explores how AI and digital technologies are changing work and organizations in healthcare. She received her B.A. in Economics and Sociology from Columbia College and PhD in management from Columbia Business School.

The NYU Stern School of Business' Center for the Future of Management (CFM) draws from expertise across NYU Stern to foster cutting-edge research on major shifts in the global business landscape, such as globalization, the changing nature of work, and opportunity and challenges from new technologies such as artificial intelligence. The Center also hosts conferences for academic, industry and policy audiences at the intersection of technology, strategy and public policy, including conferences on the Economics of Robots and Artificial Intelligence at NYU's Washington DC location, and on the Benefits and Challenges of Open Source Software at NYU Stern's NYC location.

Note

This document is approved for public dissemination. The document contains no business-proprietary or confidential information. Document contents may be reused by the government in developing the AI Action Plan and associated documents without attribution.